

ACCTG 211H — Mechanics Drill Set (Answer Key)

Week 9 — Bonds, Leases, and Covenant Slack

Drill Set

Quick, single-skill reps to build mechanical fluency before the week's first class. Graded for **completion** — work each one by hand, then check it against the key that follows and re-do any you missed. Round dollar amounts to the nearest dollar; small differences from rounding present-value factors are acceptable.

1. (*Premium or discount?*) A bond carries a 6% coupon and is issued when the market rate is 8%. Will it sell at a premium, a discount, or par? Over the bond's life, will interest expense be greater than, less than, or equal to the cash coupon?
2. (*Issue price.*) Compute the issue price of a three-year, \$100,000 bond with a 6% annual coupon when the market rate is 8%. Use annual compounding.
3. (*Effective-interest amortization.*) The bond in Drill 2 is issued at \$94,846. Under the effective-interest method, compute year-1 interest expense, the discount amortized, and the carrying value at the end of year 1.
4. (*Issuance entry.*) Record the journal entry to issue the bond in Drill 2 (use the gross method, with a separate discount account).
5. (*Bond at par.*) A \$200,000 bond with a 5% annual coupon is issued at par. Record (a) the issuance entry and (b) the first annual interest-payment entry.
6. (*Lease liability.*) A four-year lease requires payments of \$10,000 at the end of each year; the discount rate is 6%. Compute the initial lease liability (equal to the right-of-use asset).
7. (*Lease classification.*) State whether each indicates a **finance** lease under ASC 842: (a) ownership transfers to the lessee at the end of the term; (b) the lease term is nine months with no purchase option; (c) the term covers 90% of the asset's economic life; (d) the present value of payments is 97% of the asset's fair value.
8. (*Interest-coverage covenant.*) A firm has EBIT of \$4,800,000 and interest expense of \$1,200,000. A covenant requires interest coverage of at least 3.0×. Compute the ratio and the covenant slack.
9. (*Debt-to-EBITDA covenant.*) A firm has total debt of \$90 million and EBITDA of \$30 million. A covenant caps debt-to-EBITDA at 3.5×. Compute the ratio and state whether the covenant is satisfied.

Self-Check Answers

1. **Premium or discount? Discount** — the coupon (6%) is below the market rate (8%), so investors will pay less than face. Interest expense will be **greater than** the cash coupon, because expense is the carrying amount times the 8% market rate.
2. **Issue price.** Present value at 8%: coupons = $6,000 \times \text{annuity factor } (2.5771) = \$15,463$; face = $100,000 \times \text{PV factor } (0.7938) = \$79,383$. Issue price **\$94,846** (a \$5,154 discount).
3. **Effective-interest amortization.** Year-1 interest expense = $94,846 \times 8\% = \$7,588$. Cash coupon = \$6,000, so discount amortized = $7,588 - 6,000 = \$1,588$. Carrying value end of year 1 = $94,846 + 1,588 = \$96,434$.
4. **Issuance entry (gross method).** Dr Cash \$94,846; Dr Discount on Bonds Payable \$5,154; Cr Bonds Payable \$100,000.
5. **Bond at par.** (a) Dr Cash \$200,000; Cr Bonds Payable \$200,000. (b) Dr Interest Expense \$10,000 ($200,000 \times 5\%$); Cr Cash \$10,000.
6. **Lease liability.** PV of \$10,000 for four years at 6% = $10,000 \times \text{annuity factor } (3.4651) = \$34,651$.
7. **Lease classification.** (a) finance (ownership transfers); (b) not a finance lease — a short-term (12-month) lease; (c) finance (major part of economic life); (d) finance (PV is substantially all of fair value). Only (b) is not a finance lease.
8. **Interest coverage.** Ratio = $4,800,000 \div 1,200,000 = 4.0\times$. Slack = $4.0 - 3.0 = 1.0\times$ (in compliance).
9. **Debt-to-EBITDA.** Ratio = $90 \div 30 = 3.0\times$. Since $3.0\times < 3.5\times$, the covenant **is satisfied** (slack of $0.5\times$).